Overdispersion and quasilikelihood

Learning goals

Be able to explain what overdispersion is.

Understand quasilikelihood methods:

- Be able to define a quasi log-likelihood.
- Be able to compute quasi log-likelihood, scaled quasideviance, and quasideviance.
- Be able to perform a test for model selection between nested models
- Be able to estimate parameters and perform inference for quasi binomial/Poisson regressions.

Overdispersion

Data are overdispersed when the actual $\mathrm{Var}\, Y_i$ is larger than it should be, according to a model for Y_i (e.g., Poisson and binomial distributions).

For a binomial observation, if there is dependence between the trials, or if the success probability changes from trial to trial, then we can get overdispersion. Similarly, for a Poisson process dependence between events or a non-constant rate can produce overdispersion.

Quasilikelihood is one way to deal with overdispersion. Our main application of quasilikelihood theory is to generalise the binomial and Poisson regression models to allow for overdispersion.

Overdispersion

$Y_1, Y_2, \ldots, Y_n$ with $Y_i \sim \text{Poisson}(\lambda_i)$

$\Rightarrow \quad \mathbb{E}(Y_i) = \text{Var}(Y_i) = \lambda_i$

$\bar{Y} = 100 \quad < \quad s^2 = 400 \rightarrow$ Overdispersion

Evidence that the Poisson variance assumption may not be adequate.

Possible solutions:

- Using a different model for y_i

$$Y_1 - Y_n \sim NB(\lambda, \mu) \equiv \text{Poisson}(\lambda_i)$$

$$\lambda_i \sim \text{Gamma}(\gamma, \theta)$$

$$\mathbb{E}(y_i) = \mu_i$$

$$\text{Var}(y_i) = \mu_i + \alpha \mu_i^2$$

- Use ↦ quasilikelihood

Quasilikelihood

$Y_i \sim \text{Poisson}(\lambda_i)$ ⟶ We do not need to specify the full distribution of Y to model the relationship between Y and X.

- Instead we can specify only:
  - $\mathbb{E}(Y_i) = \mu_i = \lambda_i$
  - $\text{Var}(Y_i) = v(\mu_i)\phi$
  - $g(\mu_i) = x_i^t \beta$

Quasilikelihood allows us to model { mean & variance } without specifying the distribution of Y!

General idea

- Poisson model $\mapsto Var(y_i) = \mu_i = v(\mu_i) \cdot \phi$ ($\phi = 1$)
- Quasi likelihood $\mapsto Var(y_i) = \phi \mu_i$

- provides flexibility to model the variance!

$\phi > 1 \Rightarrow$ Overdispersion

Quasilikelihood

So far we have been fitting models using maximum likelihood. This has meant assuming that there is a probability distribution for the data.

Suppose that we are able to specify the link function (i.e., how the mean response is affected by several predictors) and the variance function (i.e., how the variability of the response changes with the average response) of the data, but that we do not have a strong idea about the appropriate distributional form for the response variable.

The IWLS algorithm requires only the link (g) and variance (v) functions to fit a glm and it assumes:

$$\begin{aligned} g(\mu_i) &= \mathrm{x}_i^T \boldsymbol{\beta} \\ \mathbb{E}\, Y_i &= \mu_i \\ \mathrm{Var}\, Y_i &= \phi v(\mu_i). \end{aligned}$$

The IWLS algorithm does not require distributional assumptions for the response variable.

Quasilikelihood

However, many inference procedures still need a likelihood. Can we propose a suitable substitute for a likelihood that can be computed without assuming a distribution?

Quasilikelihood is a general method for model fitting and inference that works when we do not have a likelihood. As such it is more widely applicable than maximum likelihood, however in general it provides less efficient estimators (larger variance than MLE). So if you have information about the distribution, you are advised to use it.

For the quasilikelihood, we only need the mean μ and variance $\phi v(\mu)$.

How does the quasi-likelihood works?

Let's first review how do we work when we have a log-likelihood

- When we make assumptions about the distribution of $y \longmapsto$ we can define the log-likelihood ℓ.

ℓ = log-likelihood

- To find the MLE $\longmapsto$ $U(\mu) = \left.\frac{\partial \ell}{\partial \mu}\right|_{\mu=\hat{\mu}_{MLE}} = 0$ ← Score function

Remember from MLE theory $\longmapsto$ $\mathbb{E}\,U(\mu) = 0$ & $-\mathbb{E}\left[\frac{\partial U(\mu)}{\partial \mu}\right] = \mathbb{E}\left[U(\mu)^2\right]$

Quasilikelihood Case

analogous quantity

If $\mathbb{E}\, Y_i = \mu_i$ and $\operatorname{Var} Y_i = \phi v(\mu_i)$, define the score

we define → $u_i = \dfrac{Y_i - \mu_i}{\phi v(\mu_i)}.$ → Next we see it behaves like a likelihood score!

We have

$$\mathbb{E}\, u_i = 0, \quad \operatorname{Var} u_i = \frac{1}{\phi v(\mu_i)},$$

$$-\mathbb{E}\frac{\partial u_i}{\partial \mu_i} = -\mathbb{E}\frac{-\phi v(\mu_i) - (Y_i - \mu_i)\phi v'(\mu_i)}{(\phi v(\mu_i))^2} = \frac{1}{\phi v(\mu_i)} = \mathbb{E}\, u_i^2.$$

These properties are all shared by $\frac{\partial \log L}{\partial \mu}$ and it is from these that many of the properties of MLE are derived. This suggests that we can use $\int u d\mu$ like a log-likelihood.

→ see next slide

▶ When we have a log-likelihood ℓ then ⟶

Score ⟼ $u(\mu) = \dfrac{\partial \ell(\mu)}{\partial \mu}$

We can recover $\ell(\mu)$ by integrating $u(\mu)$:

$$\ell(\mu) = \int u(\mu)\, d\mu + c$$

In the case of the quasilikelihood the idea is similar:

$$u_i = \frac{Y_i - \mu_i}{\phi v(\mu_i)}.$$

⟶ $\int u\, d\mu$ ↦ Play the role of a log-likelihood.

Quasilikelihood

Define

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi v(t)} dt \quad (\leq 0),$$

then $\frac{\partial Q_i}{\partial \mu_i} = u_i$ and the quasi log-likelihood is

$$\boxed{Q = \sum_{i=1}^{n} Q_i.}$$

For GLMs, maximizing the quasi (log) likelihood gives a consistent estimator for β. We can estimate ϕ using $X^2/(n-p)$ as before.

The usual asymptotic properties expected of maximum likelihood estimators also hold for quasi-likelihood-based estimators.

Quasilikelihood is more widely applicable than maximum likelihood, however in general it provides less efficient estimators (larger variance than MLE). So if you have information about the distribution, you are advised to use it.

Quasideviance

We can also form a quasideviance. The quasi (log) likelihood for the saturated model is clearly 0, so we get the scaled quasideviance:

$$\frac{D_Q}{\phi} = -2\left[\sum Q_i - \sum Q_i^s\right]$$
$$\overset{(*)}{=} -2\sum Q_i$$

and the quasideviance is

$$D_Q = -2\sum \int_{y_i}^{\mu_i} \frac{y_i - t}{v(t)}\,dt.$$

How far the fitted model is from the saturated model.

(*) Saturated model $\rightarrow \mu_i = y_i$

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi v(t)}\,dt \Rightarrow Q_i^s = 0$$

Scaled quasideviance for model selection (nested models)

If model A is nested within model B, under the null hypothesis that model A is correct we have

$$\begin{aligned} F_Q &= \frac{(D_Q^A - D_Q^B)/s}{\hat{\phi}} \\ &\approx F_{s,n-p}, \end{aligned}$$

where D_Q^A and D_Q^B denote quasideviances for models A and B, respectively, we have n observations, model A has $p - s$ parameters, and model B has p parameters.

- $\hat{\phi} = X^2/(n-p)$, where X^2 (Pearson's chi-squared) is calculated using model B.

Quasi Poisson

$$\operatorname{Var} Y = \phi\mu.$$

So $v(\mu) = \mu$ just as for the Poisson, but now we no longer require $\phi = 1$.

$$Q_i = \frac{1}{\phi}\left[y_i \log \mu_i - \mu_i\right]$$

which is the same as the Poisson log likelihood, except for the factor $\frac{1}{\phi}$.

Thus, IWLS will give that $\hat{\beta}$ which maximises the quasi (log) likelihood. Because IWLS does not depend on ϕ, we can fit a quasi Poisson by pretending it is a regular Poisson regression model.

- $Y \sim$ Poisson $\quad E(Y) = \lambda = \mu$

$$Var(Y) = \lambda = \mu = v(\mu)$$

$$v(\mu) = \mu, \quad \phi = 1$$

- Over dispersion

$$E(y) = \mu \quad \text{and} \quad Var(y) = \phi\mu$$

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi \underbrace{v(t)}_{=t}} dt = \frac{1}{\phi} \int_{y_i}^{\mu_i} \frac{y_i}{t} - 1 \, dt$$

$$= \frac{1}{\phi} [y_i \log t - t]_{y_i}^{\mu_i} = \frac{1}{\phi} [y_i \log \mu_i - \mu_i]$$

Quasi binomial

$$\operatorname{Var} Y = \phi m p(1-p) = \phi \mu (m-\mu)/m.$$

So $v(\mu) = \mu(m-\mu)/m$ just as for the binomial. The difference is that we no longer fix $\phi = 1$.

$$Q_i = \frac{1}{\phi}\left[y_i \log \frac{\mu_i}{m_i - \mu_i} + m_i \log(m_i - \mu_i) \right]$$

which is the same as the binomial log likelihood, except for the factor $\frac{1}{\phi}$.

So, as for the quasi Poisson, we can fit a quasi binomial using IWLS by just pretending it is a regular binomial regression model.

Overdispersion

- Poisson model $\mapsto Var(y_i) = \mu_i = v(\mu_i) \cdot \underbrace{\phi}_{=1}$ → ϕ is the dispersion parameter
- Quasi likelihood $\mapsto Var(y_i) = \phi \mu_i$
- provides flexibility to model the variance!

$\phi > 1 \Rightarrow$ Overdispersion

- Model $\hat{\beta}$ coefficients not affected
- Variances affected → $\phi > 1$

Quasi binomial and quasi Poisson

It also follows that for the quasi binomial and quasi Poisson, the deviance is the same as for the binomial and Poisson models applied to the same data. However, for the quasi binomial/Poisson models, the scaled deviance and the deviance are different.

Thus, for quasi binomial/Poisson models, we can't use the deviance to test for model adequacy, and when comparing models we can't just compare deviance and use a χ^2 test, instead we have to scale by s (the difference in df) and $\hat{\phi}$, and use an F test. In R, this can be done using the anova command (drop1 and step do not work).

Overdispersion in binomial/poisson regression

If you ignore overdispersion when fitting a binomial/poisson regression, your estimation is unaffected, but your inference changes.

With proper modelling of overdispersion,

- Your F statistic is reduced, making model comparison less significant in general (so you may end up with fewer significant variables in the model).
- Your estimate of Σ also increases, so you get larger CI for your parameter estimates.

$$\Sigma_{ii} = g'(\mu_i)^2 v(\mu_i)\phi \quad , \quad \phi > 1$$

$$\mathrm{Var}(\hat{\beta}) = (X^t \Sigma^{-1} X)^{-1} \propto \Sigma \phi$$

Example

See troutegg.pdf.

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